

overhead means that future designs must optimise layout, connectivity, syndrome extraction cycles, and decoder performance. Real-time classical decoders, including minimum-weight perfect matching and machine-learning-assisted approaches, will be essential for identifying error patterns quickly enough to preserve logical information.

Another important direction is the development of ‘fault-tolerant quantum operations’, where gates, measurements, and state preparation are executed without spreading errors uncontrollably. Techniques such as lattice surgery, magic-state distillation, transversal gates, and code deformation are expected to play a central role in scalable quantum computation. In parallel, quantum networking and modular architectures may allow multiple smaller processors to be connected through photonic links, reducing the engineering burden of building one monolithic quantum chip.

Ultimately, robust quantum systems will be built through co-design: quantum algorithms must be adapted to hardware constraints, error correction must be matched to noise models, and control systems must operate with high precision and low latency. This holistic approach will enable reliable quantum cloud platforms, secure quantum communication networks, and practical applications in chemistry, optimisation, cryptography, and materials discovery.

Best practices for building robust quantum systems

Design for fault tolerance from the architectural stage: Robust quantum systems should be planned around fault-tolerant quantum computing principles from the beginning, rather than treating error correction as an add-on layer. The physical qubit layout, coupling topology, control wiring, readout architecture, and classical feedback path must be compatible with repeated syndrome extraction, logical-qubit encoding, and scalable correction

cycles. Architectures based on surface codes, colour codes, bosonic codes, or hybrid quantum error correction schemes should be selected according to the hardware noise profile and connectivity constraints.

Maintain physical error rates below the fault-tolerance threshold: A core requirement is that single-qubit gates, two-qubit gates, idle operations, and measurement errors remain below the error threshold of the selected code. For surface-code architectures, this typically requires physical error rates well below the percent level, with practical systems aiming for much lower values to reduce logical error rates and physical-qubit overhead. Continuous improvement in gate fidelity, coherence time, leakage suppression, and readout accuracy is therefore essential for building reliable logical qubits.

Perform continuous noise characterisation and hardware-aware modelling: Quantum processors must be monitored using detailed noise diagnostics, including relaxation time, dephasing time, gate infidelity, leakage errors, crosstalk, measurement assignment errors, correlated noise, and non-Markovian behaviour. Techniques such as randomized benchmarking, gate-set tomography, cycle benchmarking, and noise spectroscopy should be used to construct accurate error models. These models allow engineers to select suitable error correction codes, optimise pulse schedules, and design compilers that avoid noisy qubit regions or error-prone gate sequences.

Optimise qubit layout, connectivity, and control routing: Scalable robustness depends heavily on physical layout. In superconducting processors, qubits should be arranged to support nearest-neighbour interactions required by surface codes while minimising unwanted coupling, frequency collisions, and microwave crosstalk. In trapped-ion or neutral-atom systems, ion shuttling, optical addressing, and reconfigurable

connectivity must be engineered to reduce control errors. The layout should also support efficient placement of data qubits, ancilla qubits, readout resonators, couplers, and cryogenic wiring without compromising thermal stability or signal integrity.

Implement fast and reliable syndrome extraction cycles: Quantum error correction depends on repeated measurement of stabilisers without directly measuring or destroying the encoded logical quantum state. Therefore, systems must support high-fidelity ancilla preparation, controlled entangling gates, rapid measurement, and reset operations. The syndrome extraction cycle must be short compared to the coherence time of physical qubits, and measurement-induced errors must be carefully controlled. Any delay in syndrome extraction increases the probability of accumulated errors and reduces the effectiveness of logical protection.

Use low-latency classical decoding and real-time feedback: A robust quantum system requires close integration between the quantum processor and classical control hardware. Syndrome data must be processed by fast decoders such as minimum-weight perfect matching, union-find decoders, tensor-network decoders, or machine-learning-assisted decoders. These decoders must operate within strict latency limits so that correction information can be applied or tracked in real time. Hardware accelerators such as FPGAs, GPUs, or cryogenic CMOS controllers may be required to support large-scale decoding for thousands or millions of physical qubits.

Minimise leakage, crosstalk, and correlated errors: Error correction codes are often designed under assumptions of mostly local and independent errors, but real quantum hardware can exhibit leakage outside the computational subspace, correlated multi-qubit errors, parasitic coupling,